RAHUL SAINI

AI Engineer — LLM Applications, RAG Pipelines & Agentic AI

Himmatpura, Bassi, Jaipur | +91 9694770048 | rahulsainihimmatpura@gmail.com | GitHub | LinkedIn | LeetCode

SUMMARY

AI Engineer who builds production-style LLM systems — from RAG pipelines to multi-agent applications — and ships them with measurable results, not just prototypes. Most recently, iterative retrieval and prompt tuning on a resume-screening service raised top-candidate ATS match accuracy from 70% to 90%. Strong foundation in Python, API development, and building systems that fail gracefully rather than crash, backed by 470+ LeetCode problems solved.

TECHNICAL SKILLS

- **Programming:** Python, C++, SQL
- **LLMs & GenAI:** LLMs, RAG, Prompt Engineering, Agentic AI, MCP, Semantic Search, Recommendation Systems, NLP
- **Frameworks:** LangChain, LangGraph, CrewAI, FastAPI, PyTorch, Pydantic, Sentence Transformers, Streamlit
- **Vector Databases:** FAISS, ChromaDB
- **Deployment & Tools:** Docker, Google Cloud Platform (GCP), Git, GitHub, MLflow, REST APIs
- **LLM Ecosystem:** OpenAI API, Gemini API, Hugging Face, Ollama, Tavily Search API

PROJECTS

Adaptive Corrective RAG (CRAG) with Knowledge Refinement

LangGraph · LangChain · FAISS · OpenAI · Tavily · Pydantic — GitHub

- Designed and implemented the Corrective-RAG architecture from the original research paper: an LLM-based retrieval evaluator scores retrieved documents and routes each query through internal retrieval, web-search fallback, or a hybrid path before generation.
- Built sentence-level knowledge refinement (decomposition + LLM relevance filtering) to strip irrelevant retrieved content, directly reducing hallucination risk in final answers.
- Added automatic query rewriting and Tavily web-search fallback for low-confidence retrievals, enforcing schema-validated (Pydantic) structured outputs for reliable routing across the LangGraph workflow. Instrumented each stage with logging and error handling so a failure degrades gracefully instead of crashing.

AI Resume Screener — Production API & Deployment

FastAPI · Streamlit · FAISS · Gemini · Docker · GCP — Live Demo | GitHub

- Designed and deployed a REST API and microservice for AI-powered resume screening, combining keyword matching with FAISS + Sentence-Transformer vector similarity for hybrid candidate ranking against a job description.
- **Improved top-candidate ATS match score from ~70% to ~90%** through iterative evaluation and tuning of the ranking logic and retrieval scoring against target JD requirements.
- Integrated the Gemini API to generate recruiter-style candidate summaries, streamed to the UI in real time via WebSockets; containerized with Docker and deployed on GCP.
- Added batch processing to screen multiple resumes per request instead of one at a time, plus structured logging and error handling across the pipeline so a single malformed resume or API failure doesn't take down the service.

Multi-Agent AI Blog Writing System

LangGraph · LangChain · GPT-4.1 Mini · Tavily · Gemini Image API · Streamlit — GitHub

- Built an orchestrator-worker multi-agent pipeline: an orchestrator agent produces a structured writing plan (title, audience, tone, section tasks), fan-out worker agents write sections in parallel, merged by a reducer into one document.
- **Improved generated content quality by ~15%** by rewriting agent prompts to enforce fact-grounding and strict instruction-following, validated through manual rubric-based evaluation of output quality.
- Integrated an adaptive router agent that selects closed-book, hybrid, or open-book generation depending on whether a topic needs external research (via Tavily); auto-generated technical diagrams via the Gemini image API. Added logging and error handling around each agent stage so a single worker failure doesn't crash the full pipeline.

OTHER PROJECTS

- Multi-Agent AI Learning System — CrewAI research/writing agents that auto-generate structured learning guides. Also built local & adaptive RAG pipelines (Ollama + ChromaDB) and an Ollama-powered Reddit sentiment/brand-monitoring dashboard.
- Hybrid Movie Recommender System — collaborative + content-based filtering re-ranked via XGBoost Learning-to-Rank; evaluated with Precision@K/Recall@K/NDCG, with all experiments logged and tracked in MLflow; deployed on GCP with Docker.

EDUCATION

Bachelor of Science (Mathematics) — Aravalli College, Bassi (2024–2025, 1 yr completed) | **Sr. Secondary (Science–Mathematics)** — Govt. Sr. Secondary School, Toonga (2023)